NAME:

SURNAME:

POLITO ID NUMBER (Matricola):

1. Which of the following is the eutectoid composition of carbon in steel?

- A) 0.02% C
- B) 0.76% C
- C) 2.14% C
- D) 4.3% C

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2. At what temperature does the eutectoid reaction occur in the iron-carbon diagram?

- A) 723°C
- B) 912°C
- C) 1147°C
- D) 1394°C

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3. Which of the following phases is present in the iron-carbon diagram at room temperature for steel with 0.2% carbon?

- A) Ferrite and Cementite
- B) Austenite
- C) Ferrite only
- D) Martensite

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4. What is the maximum solubility of carbon in ferrite (α-iron) in the iron-carbon phase diagram?

- A) 0.022%
- B) 0.76%
- C) 2.14%
- D) 6.67%

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5. Which phase exists at temperatures between 912°C and 1394°C in pure iron?

- A) Austenite (γ)
- B) Ferrite (α)
- C) Delta ferrite (δ)
- D) Cementite (Fe3C)

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6. Which of the following statements about cementite (Fe3C) is true?

- A) It is a ductile phase.
- B) It has a body-centered cubic (BCC) structure.
- C) It is a hard and brittle intermetallic compound.
- D) It is the same as ferrite.

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7. What is the percentage of carbon at the eutectic point in the iron-carbon diagram?

- A) 0.76%
- B) 2.14%
- C) 4.3%
- D) 6.67%

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8. Which phase transformation occurs during the cooling of austenite with 0.8% carbon at 723°C?

- A) Austenite to Ferrite
- B) Austenite to Pearlite
- C) Ferrite to Cementite
- D) Cementite to Austenite

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9. What type of reaction is the eutectoid reaction in the iron-carbon diagram?

- A) Liquid to Solid + Solid
- B) Solid to Solid + Solid
- C) Solid to Liquid + Liquid
- D) Liquid to Liquid + Solid

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10. Which of the following describes the structure of pearlite?

- A) Single phase with a BCC structure
- B) Alternate layers of ferrite and cementite
- C) Hard intermetallic compound with 6.67% carbon
- D) Single phase with a FCC structure
